



Installation of Jenkins

Introduction

Jenkins is a **popular open-source automation server** that helps in **Continuous Integration (CI) and Continuous Deployment (CD)**.

- How do you install Jenkins on different operating systems?
- What are the prerequisites for Jenkins installation?
- How do you start and access Jenkins after installation?

This guide provides **step-by-step installation instructions for Jenkins** on **Windows, Linux, and macOS**.

Section 1: Learn

System Requirements

Component	Minimum Requirement
RAM	1 GB (2 GB recommended)
Disk Space	50 GB (recommended for large projects)
Operating System	Windows, macOS, Linux (Ubuntu, CentOS, RedHat)
Java Version	Java 11 or later
Network Port	Default: 8080 (can be changed)

Why Install Jenkins?

- Automates software testing and deployment.
- Reduces human errors in manual deployment.
- Speeds up development cycles with CI/CD pipelines.



- Works with various DevOps tools like Docker, Kubernetes, AWS, and Terraform.

Interesting Fact

Jenkins was **originally developed in 2004** as **Hudson** and later renamed **Jenkins in 2011** after Oracle acquired Sun Microsystems.

Section 2: Practice

1. Installing Jenkins on Ubuntu

Step 1: Update System Packages

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install Java (Required for Jenkins)

```
sudo apt install openjdk-11-jdk -y
```

Verify Java installation:

```
java -version
```

Step 3: Add Jenkins Repository and Install Jenkins

```
wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo apt-key add -  
sudo sh -c 'echo deb http://pkg.jenkins.io/debian-stable binary/ >  
/etc/apt/sources.list.d/jenkins.list'  
sudo apt update  
sudo apt install jenkins -y
```



Step 4: Start and Enable Jenkins

```
sudo systemctl start jenkins  
sudo systemctl enable jenkins
```

Check Jenkins status:

```
sudo systemctl status jenkins
```

2. Installing Jenkins on Windows

Step 1: Download Jenkins Installer

- Download the **Jenkins Windows Installer** from [Jenkins Official Website](https://jenkins.io/download/windows/).

Step 2: Run the Installer

1. Open the downloaded **jenkins.msi** file.
2. Click **Next** and choose the installation directory (default is **C:\Program Files\Jenkins**).
3. Click **Install** and wait for the process to complete.

Step 3: Start Jenkins

1. Open **Windows Services** (**services.msc**).
2. Find **Jenkins** and click **Start**.

Step 4: Retrieve Initial Admin Password

1. Open **File Explorer** → Navigate to:

```
C:\Program Files\Jenkins\secrets\initialAdminPassword
```

2. Open the file and copy the password.

Jenkins is now **installed and running on Windows!**



3. Installing Jenkins on macOS

Step 1: Install Homebrew (If Not Installed)

```
/bin/bash -c "$(curl -fsSL  
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Step 2: Install Java (Required for Jenkins)

```
brew install openjdk@11
```

Step 3: Install Jenkins

```
brew install jenkins-lts
```

Step 4: Start Jenkins

```
brew services start jenkins-lts
```

Check Jenkins status:

```
brew services list
```

Jenkins is now **installed on macOS!**

4. Accessing Jenkins for the First Time

Step 1: Open Jenkins in a Web Browser

1. Open a browser and go to:

```
http://localhost:8080
```



2. If running on a remote server, replace **localhost** with the server's IP address.

Step 2: Retrieve the Initial Admin Password

For **Linux/macOS**:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

For **Windows**:

```
C:\Program Files\Jenkins\secrets\initialAdminPassword
```

Copy and paste the password into the Jenkins setup page.

Step 3: Install Suggested Plugins

1. Click **"Install Suggested Plugins"** to install common plugins automatically.
2. Wait for the installation to complete.

Step 4: Create an Admin User

1. Enter a **Username, Password, Full Name, and Email Address**.
2. Click **Save and Continue**.

Jenkins is now **ready for use**!

Section 3: Know More

Frequently Asked Questions

1. **What is the default Jenkins port?**
 - o Jenkins runs on **port 8080** by default.
2. **How do I change the Jenkins port?**



- Edit the Jenkins configuration file:

```
sudo nano /etc/default/jenkins
```

- Change `HTTP_PORT=8080` to another port, e.g., `9090`, then restart Jenkins:

```
sudo systemctl restart jenkins
```

3. What if Jenkins doesn't start after installation?

- Check logs using:

```
sudo journalctl -u jenkins -f
```

4. Where is Jenkins data stored?

- Jenkins stores data in `/var/lib/jenkins/` on Linux and `C:\ProgramData\Jenkins` on Windows.

5. How can I back up Jenkins?

- Copy the `/var/lib/jenkins` directory to a backup location.

Conclusion

Jenkins installation is the first step towards automating CI/CD workflows. Start by installing Jenkins, configuring settings, and securing access, and soon you'll be ready to build automation pipelines!